Numerical Differentiation

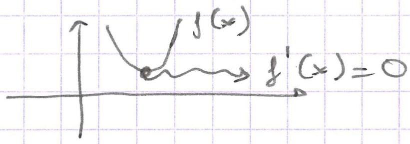
Differentiation

↳ when ?

* rate of change : eg, temperature chang, chemical concentrate speed, acc. etc.

* Taylor approximation (Topic 3)
  ↳ error analysis
  Linearization of a function about a point
  ↳ eg. dynamical systems

* Optimization ⇒ finding min / max of a functio



we will use this in Topic 5

Why numerical differentiation?

⇝ approximate $f'(x)$ when we can't determine it exactly

eg.
(1) we don't know f explicitly but have some data

y_1 y_2 y_3 - - - ↳ assume comes from some $f(x)$
x_1 x_2 x_3 - - - we want $f'(x)$

eg. experiments (measure chemical concentrate every Δt seconds)

numerically we can evaluate cost function but don't know it ↳ eg simulation

evaluating $f(x)$ is costly or introduces errors ⇝ complicated closed-form expression.

Finite Differences

Derivative definition

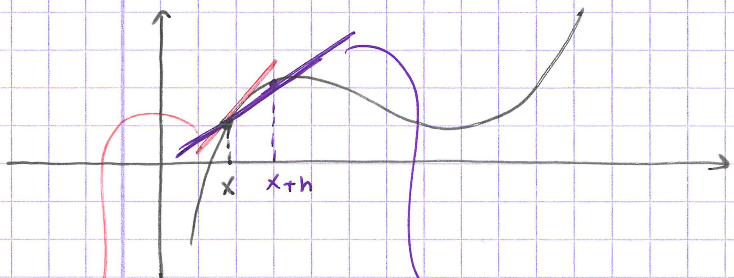
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Difference approximation

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

h → can be positive or negative.

$|h|$ is small



slope of tangent $f'(x)$

slope of the secant line → approximation $\frac{f(x+h) - f(x)}{h}$

Forward difference $f'(x) \approx \frac{f(x+h) - f(x)}{h}$

Backward difference $f'(x) \approx \frac{f(x-h) - f(x)}{-h}$

↳ h here is positive

Backward diff. error

$$f'(x) = \frac{f(x) - f(x-h)}{h} + \frac{h}{2} f''(\xi)$$

(Topic 3)

Error: By Taylor's theorem, for some $\xi \in [x, x+h]$

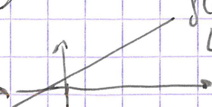
$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2} f''(\xi)$$

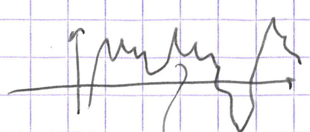
Rearrang

$$f'(x) = \underbrace{\frac{f(x+h) - f(x)}{h}}_{\text{fwd. difference}} - \frac{h}{2} f''(\xi)$$

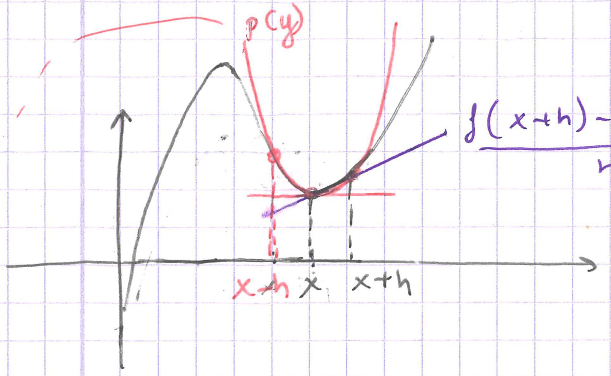
$$\left| f'(x) - \frac{f(x+h) - f(x)}{h} \right| = \frac{h}{2} f''(\xi)$$

$O(h)$





$f''(x) \neq 0$ can be very high



$\frac{f(x+h)-f(x)}{h}$ } fwd difference

⟹ linear interpolation

interpolate using a quadratic polynomial $p(y)$

$$f'(x) \approx p'(y)$$

$$p(y) = f(x) + \frac{f(x+h)-f(x-h)}{2h}(y-x)$$

$$+ \frac{f(x-h)-2f(x)+f(x+h)}{2h^2}(y-x)^2$$

$$p'(y) = \frac{f(x+h)-f(x-h)}{2h} + 2\frac{f(x-h)-2f(x)+f(x+h)}{2h^2}(y-x)$$

when $y=x$

$$p'(x) = f(x+h)-f(x-h)/2h$$

Centred-difference :

$$f'(x) \approx p'(x) = \frac{f(x+h)-f(x-h)}{2h}$$

This is a three-point formula since we are essentially using (interpolating using) three points. But, only two function evaluations are needed.

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* same formula can be obtained by using Taylor series as well.
* combine fwd & bwd differences

$f(x+h) =$

$f(x-h) =$

Error:

$$f(x+h) = f(x) + \overset{(x+h-x)}{h}\, f'(x) + h^2 f''(x)/2 + h^3 f'''(\xi_+)/6$$

$$f(x-h) = f(x) - h f'(x) + h^2 f''(x)/2 - h^3 f'''(\xi_-)/6$$

$$f(x+h) - f(x-h) = 2h f'(x) + h^3 \underbrace{\left(f'''(\xi_+) + f'''(\xi_-)\right)}/6$$

$\exists\ \xi$ s.t.

$$f'''(\xi) = \frac{f'''(\xi_+) + f'''(\xi_-)}{2}$$

$$f(x+h) - f(x-h) = 2h f'(x) + h^3 f'''(\xi)/3$$

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} - \frac{h^2}{6} f'''(\xi)$$

$\hookrightarrow O(h^2)$

Truncation and round-off error

Bound $f'''(\xi)$ s.t. $|f'''(\xi)| \leq B$

$$\left| f'(x) - \frac{f(x+h) - f(x-h)}{2h} \right| \leq \frac{h^2}{6} B$$

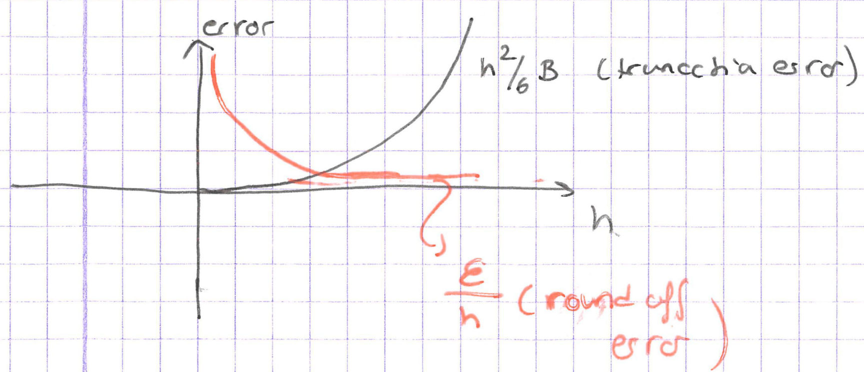
$\Rightarrow$ error due to truncation.

Unfortunately, when rounding errors are present we only have $f(x+h) + \varepsilon_1$ and $f(x-h) + \varepsilon_2$

so we get

$$\frac{f(x+h) + \varepsilon_1 - f(x-h) - \varepsilon_2}{2h} = \frac{f(x+h) - f(x-h)}{2h} + \frac{\varepsilon_1 - \varepsilon_2}{2h}$$

error due to round-off



suppose $\varepsilon_1 - \varepsilon_2$ is bounded

$$|\varepsilon_1 - \varepsilon_2| \leq \varepsilon$$

Endpoints of the domain

Determining $f(a)$ and $f(b)$ for f defined on interval $[a,b]$ we cannot use centered difference since $f(a-h)$ and $f(b+h)$ aren't defined.

Three-point forward difference

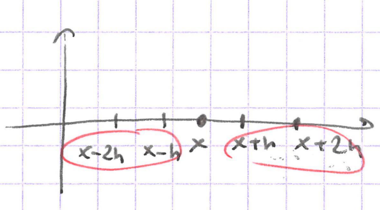
$$f'(x) \approx \frac{-f(x+2h)+4f(x+h)-3f(x)}{2h}$$

error: $$f'(x) = \frac{-f(x+2h)+4f(x+h)-3f(x)}{2h} + \frac{h^2}{3} f'''(\xi)$$

twice the error for centered difference

Five-point difference

Interpolate 5 points using a degree 4 polynomial.



odd numbers so we have equal # of nodes on both sides of x.

error: $O(h^4)$

Second derivative

Recall quadratic interpolation

$$p(y) = f(x) + \frac{f(x+h)}{2h} \quad - - -$$

$$+ \quad f(x-h) \quad - -$$

$$p'(y) = \frac{f(x+h) - f(x-h)}{2h} \quad + - - -$$

$$p''(y) = \frac{f(x-h) - 2f(x) + f(x+h)}{h^2}$$

Three-point second derivative:

$$f''(x) \approx \frac{f(x-h) - 2f(x) + f(x+h)}{h^2}$$

$\boxed{\text{Error:}}$
$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - \overbrace{f^{(4)}(\xi)\, h^2/12}^{O(h^2)}$$

Derivation (non-examinable)

$$f(x+h) = f(x) + hf'(x) + h^2 f''(x)/2 + h^3 f'''(x)/6 + - - -$$
$$f(x-h) = f(x) - hf'(x) + h^2 f''(x)/2 - h^3 f'''(x)/6 + - -$$

$$f(x+h) + f(x-h) = 2f(x) + 2h^2 f''(x)/2 + 2h^4 f^{(4)}(x)/24 + O(h^5)$$

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - \underbrace{\frac{\cancel{h^4}\, h^2}{\cancel{h^2}} f^{(4)}(x)/12 + O(h^5)}_{}$$

from Taylor's thm

$\exists\ \xi$ s.t.

$$'' \quad = \quad '' \quad - f^{(4)}(\xi)\, h^2/2$$